Lab 11 practice: surface integrals and Stokes' theorem (Zill §9.13, 9.14)

Prep for the Wed Nov 25 lab quiz. Closed book, about 10 minutes per problem. Problems 1 to 4 are surface area and surface integrals of scalar functions (graphs first, then parametric surfaces), problems 5 and 6 are flux integrals, problem 7 verifies Stokes' theorem by computing both sides, and problem 8 uses Stokes' theorem to swap a curved surface for a flat one with the same boundary. Every answer was recomputed by a second route (`lab11_check.py`).

Scalar surface integrals. For S the graph $z = g(x, y)$ over a region R of the xy-plane:

$$dS = \sqrt{1 + g_x^2 + g_y^2}\, dA, \qquad \iint_S f\, dS = \iint_R f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2}\, dA$$

For a parametric surface $\mathbf{r}(u, v)$ over a parameter region D:

$$dS = |\mathbf{r}_u \times \mathbf{r}_v|\, dA, \qquad \iint_S f\, dS = \iint_D f(\mathbf{r}(u, v))\, |\mathbf{r}_u \times \mathbf{r}_v|\, dA$$

Surface area is the case $f = 1$. Do not forget the factor $|\mathbf{r}_u \times \mathbf{r}_v|$ or the square root.

Flux. The flux of $\mathbf{F}$ through an oriented surface S with unit normal $\mathbf{n}$ is $\displaystyle\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$. State which side $\mathbf{n}$ points to before computing.

- Graph $z = g(x, y)$, **upward** normal: $\mathbf{n}\, dS = \langle -g_x, -g_y, 1 \rangle\, dA$. Downward is the negative.
- Parametric: $\mathbf{n}\, dS = \pm\, (\mathbf{r}_u \times \mathbf{r}_v)\, dA$, with the sign chosen so the normal points the way you were told. Check one point.

Stokes' theorem. Let S be a piecewise smooth oriented surface whose boundary C is a simple closed piecewise smooth curve, and let $\mathbf{F}$ have continuous first partials on an open region containing S. Then

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

Orientation of C by the right-hand rule: with the thumb along $\mathbf{n}$, the fingers curl in the direction of travel on C. For an upward normal this means counterclockwise seen from above.

Problem 1

Find the area of the part of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 4$.

Solution: area of a graph

The surface is the graph $z = g(x, y) = x^2 + y^2$, with $g_x = 2x$ and $g_y = 2y$. The plane $z = 4$ meets it where $x^2 + y^2 = 4$, so the surface lies over the disk $R : x^2 + y^2 \le 4$.

$$dS = \sqrt{1 + 4x^2 + 4y^2}\, dA$$

Switch to polar coordinates, $dA = r\, dr\, d\theta$, $0 \le r \le 2$, $0 \le \theta \le 2\pi$:

$$A = \int_0^{2\pi} \int_0^2 \sqrt{1 + 4r^2}\, r\, dr\, d\theta$$

Let $u = 1 + 4r^2$, so $du = 8r\, dr$. The limits $r = 0, 2$ become $u = 1, 17$.

$$\int_0^2 \sqrt{1 + 4r^2}\, r\, dr = \frac{1}{8} \int_1^{17} u^{1/2}\, du = \frac{1}{8} \cdot \frac{2}{3}\Big(17^{3/2} - 1\Big) = \frac{17\sqrt{17} - 1}{12}$$

$$\boxed{A = 2\pi \cdot \frac{17\sqrt{17} - 1}{12} = \frac{\pi}{6}(17\sqrt{17} - 1) \approx 36.18}$$

Sanity check: the flat disk under the surface has area $4\pi \approx 12.57$, and the paraboloid is steep near the rim, so a surface area near three times that is reasonable.

Problem 2

Evaluate $\iint_S (x^2 + y^2)\, dS$, where S is the part of the cone $z = \sqrt{x^2 + y^2}$ with $1 \leq z \leq 2$.

Solution: scalar integral over a graph

S is the graph of $g = \sqrt{x^2+y^2}$ over the annulus $R : 1 \le x^2+y^2 \le 4$. Away from the origin g is smooth, and the origin is not on S.

$$g_x = \frac{x}{\sqrt{x^2+y^2}}, \quad g_y = \frac{y}{\sqrt{x^2+y^2}}, \quad g_x^2 + g_y^2 = 1 \implies dS = \sqrt{2}\, dA$$

The integrand $f = x^2+y^2$ does not involve z, so nothing needs substituting. In polar coordinates:

$$\iint_S (x^2+y^2)\, dS = \sqrt{2} \int_0^{2\pi} \int_1^2 r^2 \cdot r\, dr\, d\theta = \sqrt{2} \cdot 2\pi \cdot \frac{2^4 - 1^4}{4} = \sqrt{2} \cdot 2\pi \cdot \frac{15}{4}$$

$$\boxed{\iint_S (x^2+y^2)\, dS = \frac{15\sqrt{2}\,\pi}{2} \approx 33.32}$$

The $\sqrt{2}$ is the slope factor of a 45° cone: the cone is steeper than its shadow by exactly $1/\cos 45^\circ$.

Problem 3

The helicoid S is parametrized by $\mathbf{r}(u, v) = \langle u\cos v,\ u\sin v,\ v\rangle$ for $0 \le u \le 1$, $0 \le v \le \pi$. Find its surface area.

Solution: area of a parametric surface

The surface winds upward like a spiral ramp, so it is not a graph over a single region of the xy-plane in any convenient way. Use the parametrization.

$$\mathbf{r}_u = \langle \cos v,\ \sin v,\ 0 \rangle, \qquad \mathbf{r}_v = \langle -u\sin v,\ u\cos v,\ 1 \rangle$$

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos v & \sin v & 0 \\ -u\sin v & u\cos v & 1 \end{vmatrix} = \langle \sin v,\ -\cos v,\ u\cos^2 v + u\sin^2 v \rangle = \langle \sin v,\ -\cos v,\ u \rangle$$

$$|\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{\sin^2 v + \cos^2 v + u^2} = \sqrt{1+u^2}$$

The parameter region is the rectangle $0 \le u \le 1$, $0 \le v \le \pi$, so

$$A = \int_0^{\pi}\int_0^1 \sqrt{1+u^2}\, du\, dv = \pi \int_0^1 \sqrt{1+u^2}\, du$$

Use $\displaystyle\int \sqrt{1+u^2}\, du = \tfrac{1}{2}\Big[u\sqrt{1+u^2} + \ln\left(u + \sqrt{1+u^2}\right)\Big]$ (trigonometric substitution $u = \tan t$):

$$\int_0^1 \sqrt{1+u^2}\, du = \frac{1}{2}\Big[\sqrt{2} + \ln\left(1+\sqrt{2}\right)\Big]$$

$$\boxed{A = \frac{\pi}{2}\Big[\sqrt{2} + \ln\left(1+\sqrt{2}\right)\Big] \approx 3.608}$$

Sanity check: the shadow of the surface on the xy-plane is a half-disk of radius 1, area $\pi/2 \approx 1.571$. The ramp is tilted, so its area must be larger than that, and 3.608 is.

Problem 4

Evaluate $\iint_S x^2 z \, dS$, where S is the cylinder $x^2 + y^2 = 4$ with $0 \leq z \leq 3$ (the curved side only, no end caps).

Solution: scalar integral where the surface is not a graph over the xy-plane

The cylinder is not a graph $z = g(x, y)$, since many points share one (x, y). Parametrize it by the angle and the height:

$$\mathbf{r}(\theta, z) = \langle 2\cos\theta,\ 2\sin\theta,\ z\rangle, \qquad 0 \le \theta \le 2\pi,\ \ 0 \le z \le 3$$

$$\mathbf{r}_\theta = \langle -2\sin\theta,\ 2\cos\theta,\ 0\rangle, \qquad \mathbf{r}_z = \langle 0,\ 0,\ 1\rangle$$

$$\mathbf{r}_\theta \times \mathbf{r}_z = \langle 2\cos\theta,\ 2\sin\theta,\ 0\rangle, \qquad |\mathbf{r}_\theta \times \mathbf{r}_z| = 2$$

So $dS = 2\,d\theta\,dz$, the circumference element $2\,d\theta$ times the height element. On S, $x^2 z = 4\cos^2\theta\ z$:

$$\iint_S x^2 z\,dS = \int_0^3 \int_0^{2\pi} 4\cos^2\theta\ z \cdot 2\ d\theta\,dz = 8\Big(\int_0^{2\pi} \cos^2\theta\,d\theta\Big)\Big(\int_0^3 z\,dz\Big) = 8 \cdot \pi \cdot \frac{9}{2}$$

$$\boxed{\iint_S x^2 z\,dS = 36\pi \approx 113.1}$$

Problem 5

Let S be the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the xy-plane, oriented **upward** (the normal has a positive **k** component). Find the flux of $\mathbf{F} = \langle x,\ y,\ z \rangle$ through S.

Solution: flux through a graph, upward normal

Orientation: upward. The surface is the graph of $g = 4 - x^2 - y^2$, with $g_x = -2x$, $g_y = -2y$, over the disk $R : x^2 + y^2 \le 4$ (where $z \ge 0$). The upward normal satisfies

$$\mathbf{n}\, dS = \langle -g_x,\ -g_y,\ 1\rangle\, dA = \langle 2x,\ 2y,\ 1\rangle\, dA$$

Its $\mathbf{k}$ component is $+1$, so this is the upward side. Put $z = 4 - x^2 - y^2$ into $\mathbf{F}$:

$$\mathbf{F} \cdot \mathbf{n}\, dS = (2x^2 + 2y^2 + 4 - x^2 - y^2)\, dA = (x^2 + y^2 + 4)\, dA$$

$$\text{Flux} = \int_0^{2\pi} \int_0^2 (r^2 + 4)\, r\, dr\, d\theta = 2\pi \Big[\frac{r^4}{4} + 2r^2 \Big]_0^2 = 2\pi(4 + 8)$$

$$\boxed{\text{Flux} = 24\pi \approx 75.40 \quad \text{(upward)}}$$

Positive flux is what you expect: $\mathbf{F}$ points away from the origin, and the upward side of the cap faces away from the origin. Reversing the orientation to downward would give -24π.

Problem 6

Let S be the sphere $x^2 + y^2 + z^2 = 4$, oriented **outward**. Find the flux of $\mathbf{F} = \langle x,\ y,\ z^2 \rangle$ through S.

Solution: flux through a closed surface, parametric

Orientation: outward, away from the origin. Parametrize with the spherical angles (φ measured from the positive z-axis):

$$\mathbf{r}(\varphi,\theta) = \langle 2\sin\varphi\cos\theta,\ 2\sin\varphi\sin\theta,\ 2\cos\varphi\rangle, \qquad 0 \le \varphi \le \pi,\ \ 0 \le \theta \le 2\pi$$

$$\mathbf{r}_\varphi = \langle 2\cos\varphi\cos\theta,\ 2\cos\varphi\sin\theta,\ -2\sin\varphi\rangle, \qquad \mathbf{r}_\theta = \langle -2\sin\varphi\sin\theta,\ 2\sin\varphi\cos\theta,\ 0\rangle$$

$$\mathbf{r}_\varphi \times \mathbf{r}_\theta = 4\sin\varphi\,\langle \sin\varphi\cos\theta,\ \sin\varphi\sin\theta,\ \cos\varphi\rangle$$

Direction check: this is a positive multiple of $\mathbf{r}$ itself for $0 < \varphi < \pi$, so it points outward and no sign flip is needed. On S, $\mathbf{F} = \langle 2\sin\varphi\cos\theta,\ 2\sin\varphi\sin\theta,\ 4\cos^2\varphi\rangle$, so

$$\mathbf{F}\cdot(\mathbf{r}_\varphi \times \mathbf{r}_\theta) = 4\sin\varphi\Big[2\sin^2\varphi\cos^2\theta + 2\sin^2\varphi\sin^2\theta + 4\cos^3\varphi\Big] = 8\sin^3\varphi + 16\sin\varphi\cos^3\varphi$$

$$\text{Flux} = \int_0^{2\pi}\int_0^{\pi} \left(8\sin^3\varphi + 16\sin\varphi\cos^3\varphi\right) d\varphi\, d\theta$$

With $\int_0^\pi \sin^3\varphi\, d\varphi = \frac{4}{3}$ and $\int_0^\pi \sin\varphi\cos^3\varphi\, d\varphi = \Big[-\frac{\cos^4\varphi}{4}\Big]_0^\pi = 0$:

$$\text{Flux} = 2\pi\Big(8\cdot\frac{4}{3} + 0\Big)$$

$$\boxed{\text{Flux} = \frac{64\pi}{3} \approx 67.02 \quad \text{(outward)}}$$

The z^2 component contributes nothing: $F_z n_z = z^2 \cdot z/2$ is odd in z, so the upper and lower halves of the sphere cancel. All of the flux comes from the $\langle x, y\rangle$ part.

Problem 7

Let $\mathbf{F} = \langle y^2 + z,\ x,\ z^2 \rangle$ and let S be the part of the paraboloid $z = 4 - x^2 - y^2$ with $z \geq 0$, oriented upward. Let C be its boundary. Verify Stokes' theorem by computing both $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS$ and $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

Solution: both sides of Stokes' theorem

Hypotheses. S is a smooth surface (a graph of a polynomial), oriented upward. Its boundary is the circle C: $x^2 + y^2 = 4$, $z = 0$, a simple closed smooth curve. $\mathbf{F}$ is polynomial, so it has continuous first partials on all of $\mathbb{R}^3$. Stokes' theorem applies.

Orientation of C. Upward $\mathbf{n}$ and the right-hand rule give counterclockwise travel on C seen from above: $\mathbf{r}(t) = \langle 2\cos t,\ 2\sin t,\ 0\rangle$, $0 \le t \le 2\pi$.

Surface side. With $P = y^2 + z$, $Q = x$, $R = z^2$:

$$\nabla \times \mathbf{F} = \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y\rangle = \langle 0,\ 1,\ 1 - 2y\rangle$$

For the graph $g = 4 - x^2 - y^2$, the upward $\mathbf{n}\,dS = \langle 2x,\ 2y,\ 1\rangle\,dA$, so

$$(\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = (0 + 2y + 1 - 2y)\,dA = dA$$

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = \iint_{x^2+y^2\le 4} 1\,dA = 4\pi$$

Line side. On C, $z = 0$ and $dz = 0$, so $\mathbf{F} \cdot d\mathbf{r} = y^2\,dx + x\,dy$. With $x = 2\cos t$, $y = 2\sin t$, $dx = -2\sin t\,dt$, $dy = 2\cos t\,dt$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \Big[-8\sin^3 t + 4\cos^2 t \Big]\,dt = 0 + 4\pi$$

since $\int_0^{2\pi} \sin^3 t\,dt = 0$ and $\int_0^{2\pi} \cos^2 t\,dt = \pi$.

$$\boxed{\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = 4\pi = \oint_C \mathbf{F} \cdot d\mathbf{r}}$$

Both sides agree. Had C been traversed clockwise, the line integral would have come out as -4π, which is how a wrong orientation shows up.

Problem 8

Let $\mathbf{F} = \langle -yz,\ xz,\ xy \rangle$ and let S be the cap of the sphere $x^2 + y^2 + z^2 = 25$ with $z \geq 3$, oriented outward (away from the origin). Find $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS$, using Stokes' theorem to replace S by a flat surface.

Solution: swap the curved surface for a disk

Boundary. The sphere meets the plane $z = 3$ where $x^2 + y^2 = 16$. So C is the circle of radius 4 centered on the z-axis at height $z = 3$. The outward normal on the cap has a positive $\mathbf{k}$ component, so by the right-hand rule C runs counterclockwise seen from above.

Hypotheses. $\mathbf{F}$ is polynomial, so it is C^1 on all of $\mathbb{R}^3$. The cap is smooth and C is a simple closed smooth curve. Stokes' theorem applies to S, and equally to the flat disk D in the plane $z = 3$ bounded by the same circle C, oriented upward so that C keeps the same direction. Since $\nabla \times \mathbf{F}$ is continuous on the solid region between S and D as well, the two surface integrals are equal to the same line integral.

Curl. With $P = -yz$, $Q = xz$, $R = xy$:

$$\nabla \times \mathbf{F} = \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y \rangle = \langle x - x,\ -y - y,\ z + z \rangle = \langle 0,\ -2y,\ 2z \rangle$$

Swap to the disk. On D, $\mathbf{n} = \mathbf{k}$ and $dS = dA$, so only the $\mathbf{k}$ component of the curl matters, and $z = 3$ there:

$$(\nabla \times \mathbf{F}) \cdot \mathbf{n} = 2z = 6$$

$$\iint_D 6\, dA = 6 \cdot \pi \cdot 4^2 = 96\pi$$

$$\boxed{\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dA = 96\pi \approx 301.6}$$

Check on the line integral. On C, $\mathbf{r} = \langle 4\cos t,\ 4\sin t,\ 3 \rangle$, and $\mathbf{F} \cdot d\mathbf{r} = z\,(x\,dy - y\,dx) = 3 \cdot 16\, dt$, so $\oint_C \mathbf{F} \cdot d\mathbf{r} = 48 \cdot 2\pi = 96\pi$.

Doing it directly on the cap is possible, with $\mathbf{n} = \frac{1}{5}\langle x, y, z \rangle$ and $(\nabla \times \mathbf{F}) \cdot \mathbf{n} = \frac{2}{5}(z^2 - y^2)$ integrated over spherical angles, but it takes far longer and returns the same 96π. The flat disk replaces a trigonometric integral over a curved cap with one line of arithmetic.